Z-residuals for Diagnosing Bayesian Models

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Section 1

Introduction

Introduction

- **Residual diagnostics** or more generally model diagnostics plays very important roles in frequentist statistics:
  - **Visualize** the residuals, for example with scatterplot and QQ plot, for identifying the potential misspecification.
  - **Quantify** the good fits of a fitted model using overall goodness-of-fit (GOF) and other diagnostic tests.
  - **Discover** the problems of a fitted model, such as family misspecification, non-linear covariate effect, outliers, and many more.
  - Identify **implementation** errors on computers (eg, wrong variable type, wrong data manipulation, computational errors like overflow/underflow/rounding errors)
- Goals of model diagnostics: 1) suggest ways to improve a model/implementation; 2) the confirmation of the good fit of a model.

Introduction (cont'd)

- Model Comparison VS Model Checking/Diagnostics
  Information criteria can compare the goodness of fits of a set of models, but *possibly all of them are poor for the datasets*.
- Residual diagnostics is not typically conducted in Bayesian statistics
- In the past few years, we have developed Z-residuals for various frequentist models, such as survival models and count regression models, and demonstrated the effectiveness of Z-residuals.
- In this talk, we show how to define Z-residuals for diagnosing Bayesian models using Bayesian hurdle models as an example.

Section 2

Hurdle Models

Hurdle Models

- Hurdle models can be viewed as a two-component mixture model consisting of
  1. a point mass at zero ($\delta_0(\cdot)$), and
  2. a zero-truncated count distribution for the positive observations.
- The probability mass function (PMF) of hurdle models is defined as

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^o, & y_i = 0, \\ (1 - \pi_i^o)\frac{p(y_i;\mu_i)}{1-p(y_i=0;\mu_i)}, & y_i > 0, \end{cases} \tag{1}$$

where π_i^o is the proportion of zeros, $p(y_i; \mu_i)$ represents a PMF for a regular count distribution with a vector of parameters μ_i, and $p(y_i = 0; \mu_i)$ is the distribution evaluated at zero.

Examples of Hurdle Models

- Hurdle negative binomial (HNB) models are defined with the following PMF:

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^o, & y_i = 0, \\ \frac{1-\pi_i^o}{1-\left(\frac{k}{\mu_i+k}\right)^k} \frac{\Gamma(y_i+k)}{\Gamma(k)y_i!} \left(\frac{\mu_i}{\mu_i+k}\right)^{y_i} \left(\frac{k}{\mu_i+k}\right)^k, & y_i > 0, \end{cases} \quad (2)$$

  where μ_i is the mean parameter, k is the dispersion parameter of NB distribution.

- Hurdle Poisson models:

$$Pr(Y_i = y_i) = \begin{cases} \pi_i^o, & y_i = 0, \\ (1 - \pi_i^o) \frac{e^{-\mu_i} \mu_i^{y_i} / y_i!}{1-e^{-\mu_i}}, & y_i > 0, \end{cases} \quad (3)$$

  where μ_i is the mean parameter of Poisson distribution.

Hurdle Models (cont'd)

- The parameters of zero and count components are linked to covariates:
  - Count model: $\log(\mu_i) = \boldsymbol{x}_i^T \boldsymbol{\alpha}$
  - Zero model: $\text{logit}(\pi_i^o) = \boldsymbol{z}_i^T \boldsymbol{\beta}$ where
  - μ_i is the mean,
  - π_i^o is the probability of zeros,
  - $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are regression coefficients for the covariates $\boldsymbol{x}_i$ and $\boldsymbol{z}_i$.

Section 3

Review of Posterior Predictive Checks

General Description of Posterior Predictive P-value (PPP)

- **What is it?** The Posterior Predictive p-value (PPP) is a goodness-of-fit measure to check if a model is consistent with the observed data, denoted by $\boldsymbol{y}^{\text{obs}}$.
- **Idea:** We compare the observed data to replicated data generated from the posterior predictive model. If the model fits well, the replicated data should look similar to the observed data.
- Define a **discrepancy statistic**, $T(\boldsymbol{y}, \boldsymbol{\theta})$, to summarize a specific discrepancy of the data, y.This can be a function of the data alone (e.g., sample mean, $T(\boldsymbol{y})$) or also depend on the model parameters, $\boldsymbol{\theta}$.
- **Posterior Predictive P-value (PPP):** It is the probability that $T(\boldsymbol{y}, \boldsymbol{\theta})$ computed on replicated data (y_{rep}) is more extreme than $T(\boldsymbol{y}, \boldsymbol{\theta})$ computed on the observed data (y_{obs}).

$$\text{PPP} = P(T(\boldsymbol{y}^{\text{rep}}, \boldsymbol{\theta}) > T(\boldsymbol{y}^{\text{obs}}, \boldsymbol{\theta}) \mid \boldsymbol{y}^{\text{obs}})$$

Computing Procedure

. . .

Posterior Samples

$\{\boldsymbol{\theta}_s\}_{s=1}^{S}$

. . .

1a: Compute **Replicated** Statistic

T_s^{rep} with $\boldsymbol{y}_s^{\text{rep}} \sim p(\boldsymbol{y} \mid \boldsymbol{\theta}_s))$

1b: Compute **Observed** Statistic

T_s^{obs} with the actual data $\boldsymbol{y}^{\text{obs}}$)

Collect

Collect

2. Aggregate & Compare All S Results

$\widehat{\text{PPP}} = \frac{1}{S}\sum_{s=1}^{S} I(T_s^{\text{rep}} \geq T_s^{\text{obs}})$

Criticisms on Posterior Predictive Checks

Difficulty in Choosing Discrepancy Statistics

- The choice of a test statistic $T(\boldsymbol{y}, \boldsymbol{\theta})$ is subjective, depending on what possible problems in the model that one suspects to exist.
- A model might pass checks with simple statistics (e.g., mean, variance) yet fail to capture other critical features of the data. This can create a false acceptance of "good" models. A "good" p-value for one statistic can mask other significant misfit *that is unknown to the modeler*.

Non-uniformity of Posterior Predictive P-values

- The data is used twice: 1) to fit the model, and 2) to test the model. The posterior predictive p-values are non-uniformly distributed with a bias towards 0.5.
- The bias results in a conservative test, which may have low statistical powers to detect genuine model misspecifications.

Section 4

Z-residuals

Randomized Predictive P-values (RSP) given True Parameters

- The postulated model for $y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}$:
  - The PMF is $p_i(y_i \mid \boldsymbol{\theta})$
  - The survival function is $S_i(y_i \mid \boldsymbol{\theta})$
- Predictive p-values for an observed y_i:
  - The **middle-value predictive p-value (MSP)**:

$$\text{msp}_i(y_i \mid \boldsymbol{\theta}) = S_i(y_i \mid \boldsymbol{\theta}) + 0.5 \times p_i(y_i \mid \boldsymbol{\theta}). \tag{4}$$

  - The **randomized predictive p-value (RSP)** is defined as:

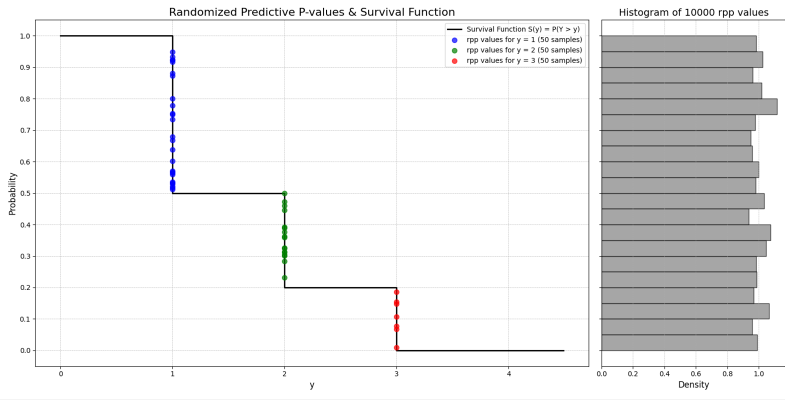
$$\text{rsp}_i(y_i \mid \boldsymbol{\theta}) = S_i(y_i \mid \boldsymbol{\theta}) + U_i \times p_i(y_i \mid \boldsymbol{\theta}) \tag{5}$$

  where $U_i \sim \text{Uniform}(0, 1)$.
- Uniformity of RSP under the true model,

$$\text{rsp}_i(y_i \mid \boldsymbol{\theta}) \mid \boldsymbol{x}_i \sim \text{Uniform}(0, 1) \tag{6}$$

Illustration of the Uniformity of RSPs

Illustration for a single y_i with fixed $\boldsymbol{x}_i$:



Proof: The uniformity of RSPs can be easily proved with elementary probability theory provided that $S_i(y_i \mid \boldsymbol{\theta})$ and $p_i 8(y_i \mid \boldsymbol{\theta})$ are the true models.

Transforming RSPs to Normally Distributed Z-residuals

- To compute the **Z-residuals**, we can transform $\mathrm{rsp}_i(y_i \mid \boldsymbol{\theta})$ with the negative normal quantile:

$$z_i^{\mathrm{rsp}}(y_i \mid \boldsymbol{\theta}) = -\Phi^{-1}(\mathrm{rsp}_i(y_i \mid \boldsymbol{\theta})), \tag{7}$$

  where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of a standard normal distribution.

- Under the true model, the Z-residual is distributed as the standard normal.

$$z_i^{\mathrm{rsp}}(y_i \mid \boldsymbol{\theta}) \mid \boldsymbol{x}_i \sim N(0, 1) \tag{8}$$

Posterior Z-residuals

- The **posterior RSP** $\text{rsp}_i^{\text{post}}(y_i)$ is defined as

$$\text{rsp}_i^{\text{post}}(y_i) = \int \text{rsp}_i(y_i \mid \boldsymbol{\theta}) P(\boldsymbol{\theta} \mid \mathbf{y}) d\boldsymbol{\theta} \tag{9}$$

- The posterior RSP is computed given the Markov Chain Monte Carlo (MCMC) samples, and denoted by $\widehat{\text{rsp}_i}^{\text{post}}(y_i)$:

$$\widehat{\text{rsp}_i}^{\text{post}}(y_i) = \frac{\sum_{t=1}^{T} \text{rsp}_i(y_i \mid \boldsymbol{\theta}^{(t)})}{T}. \tag{10}$$

- The **posterior RSP Z-residual** can be defined as

$$z_i^{\text{post-rsp}}(y_i) = -\Phi^{-1}(\widehat{\text{rsp}}^{\text{post}}(y_i)). \tag{11}$$

- The posterior MSP and posterior MSP Z-residuals can be defined in the same way.
- We will use ‘post’ to abbreviate the term ‘posterior’.

Cross-validatory RSP

- The **cross-validatory RSP**, $\text{rsp}_i^{\text{CV}}(y_i)$, is defined as

$$\begin{aligned}
\text{rsp}_i^{\text{CV}}(y_i) &= \int \text{rsp}(y_i \mid \boldsymbol{\theta}) P(\boldsymbol{\theta} \mid \boldsymbol{y}_{-i}) d\boldsymbol{\theta} && (12) \\
&= \int \big(S_i(y_i \mid \boldsymbol{\theta}) + U_i \times p_i(y_i \mid \boldsymbol{\theta})\big) P(\boldsymbol{\theta} \mid \boldsymbol{y}_{-i}) d\boldsymbol{\theta} && (13) \\
&= S_i(y_i \mid \boldsymbol{y}_{-i}) + U_i \times p_i(y_i \mid \boldsymbol{y}_{-i}) && (14)
\end{aligned}$$

- **Uniformity of CV RSP**

  It can be seen easily that

$$\text{rsp}_i^{\text{CV}}(y_i) \mid \boldsymbol{x}_i \sim \text{Uniform}(0, 1) \tag{15}$$

  as long as the prior for $\boldsymbol{\theta}$ and the distribution for $y_i \mid \boldsymbol{\theta}$ are both correctly specified.
- The proof is the same as that for the uniformity of $\text{rsp}_i(y_i \mid \boldsymbol{\theta})$.

Importance-sampling CV Z-residuals

- The ISCV RSP is computed given the MCMC samples $\{(\boldsymbol{\theta}^{(t)}); t = 1, \ldots, T\}$, and denoted by:

$$\widehat{\text{rsp}_i}^{\text{iscv}}(y_i) = \frac{\sum_{t=1}^{T}\left[\text{rsp}_i(y_i \mid \boldsymbol{\theta}^{(t)}) w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})\right]/T}{\sum_t^T w_i^{\text{IS}}(\boldsymbol{\theta}^{(t)})/T}, \tag{16}$$

where $w_i^{\text{IS}}(\boldsymbol{\theta})$ is the ratio:

$$w_i^{\text{IS}}(\boldsymbol{\theta}) = \frac{P_{\text{post(-i)}}(\boldsymbol{\theta} \mid y_{-i})}{P_{\text{post}}(\boldsymbol{\theta} \mid y_{1:n})} \times \frac{C_2}{C_1} = \frac{1}{P_y(y_j \mid \boldsymbol{\theta})} \tag{17}$$

- The **ISCV RSP Z-residual** can be defined as

$$z_i^{\text{iscv-rsp}}(y_i) = -\Phi^{-1}(\widehat{\text{rsp}_i}^{\text{iscv}}(y_i)). \tag{18}$$

- The ISCV MSP Z-residuals is calculated similarly.

Summary of Z-residual Diagnostic Methods

- Diagnostics based on Z-residuals
  - Four Z-residuals

	RSP	MSP
Post.	$z_i^{\text{post-rsp}}(y_i) = -\Phi^{-1}(\widehat{\text{rsp}_i}^{\text{post}}(y_i))$	$z_i^{\text{post-msp}}(y_i) = -\Phi^{-1}(\widehat{\text{msp}_i}^{\text{post}}(y_i))$
ISCV	$z_i^{\text{iscv-rsp}}(y_i) = -\Phi^{-1}(\widehat{\text{rsp}_i}^{\text{iscv}}(y_i))$	$z_i^{\text{iscv-msp}}(y_i) = -\Phi^{-1}(\widehat{\text{msp}_i}^{\text{iscv}}(y_i))$

  - Applying various tests to Z-residuals: Shapiro-Wilks Tests, AOV test, Bartlett test, etc. These test methods are denoted by $Z-$test, where the test indicates the test methods.

Summary of Posterior Predictive Check Methods

- Posterior predictive checks with various choice of discrepancy measures, eg.
  - $\text{PPC}_{y-\chi^2}$:

$$T(\boldsymbol{y}, \boldsymbol{\theta}) = \sum_{i=1}^{n} \frac{(y_i - E[y_i \mid \boldsymbol{\theta}])^2}{\text{Var}(y_i \mid \boldsymbol{\theta})}.$$

    We denote this method by $\text{PPC}_{y-\chi^2}$ since we apply χ^2 test to y_i directly.
  - Statistical tests (eg. Shapiro-Wilk (SW) tests) applied to point-wise Z-residuals given each $\boldsymbol{\theta}^{(t)}$ MCMC sample:

$$z_i(y_i, \boldsymbol{\theta}^{(t)}) \text{ for } i = 1, \ldots, n.$$

    Depending on what tests applied to the point-wise Z-residuals, we denote these test methods by $\text{PPC}_{Z-\text{test}}$ where the test may be SW, AOV or others.

Comparison of PPC and Z-residual Diagnostics

- Posterior predictive checks VS Z-residual diagnostics:
  - PPC involve computing point-wise

    $$z_i^{\text{rsp}}(y_i \mid \boldsymbol{\theta}^{(t)}) \text{ for } i = 1, \ldots, n$$

    for each $\boldsymbol{\theta}^{(t)}$ in the MCMC sample, applying a test method to the above point-wise z-residuals, and finally summarizing the p-values over $\boldsymbol{\theta}^{(t)}$ into a single PPC p-value.
  - Z-residual diagnostics involve summarizing over MCMC sample $\boldsymbol{\theta}^{(t)}$ for obtaining the posterior or ISCV $\text{rsp}_i(y_i)$, converting them to Z-residuals:

    $$z_i^{\text{rsp}}(y_i), \text{ for } i = 1, \ldots, n,$$

    and finally applying diagnostic methods (eg. SW test) to the above Z-residuals.

Section 5

Simulation Study I:
Detection of Mis-specified Distribution Family

Subsection 1

Data Generation and Model Fitting

Data Generation from HNB

The data is generated as follows:

- A covariate: $X_i \sim N(0, 1)$.
- Five noises: $Z_{i,j} \sim N(0, 1), j = 1, \ldots, 5$.
- Zero model: $\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_i$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 X_i$.
- The zero indicator: $O_i \sim \text{Binom}(1, \pi_i^o)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
  where $C_i \sim \text{Truncated-NB}(\mu_i, \gamma, \pi_i), \gamma = 4$ is the dispersion parameter,
  and $\pi_i = \gamma/(\gamma + \mu_i)$ is the probability.
- Regression coefficients: $\alpha_0 = 2, \alpha_1 = 8, \beta_0 = -1, \beta_1 = -1$.
- The above steps are repeated for all simulated datasets. In total, we generate 600 response variables from the HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$ and 60%.

Fitting HNB and NB Models

We fit two candidate models to the simulated datasets utilizing the function `brm()` from R package `brms`. The models are fitted as listed below:

- **HNB Model (true model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, \pi_i^o),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^{5} a_j Z_{i,j},$$

$$\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_i + \sum_{j=1}^{5} b_j Z_{i,j};$$

- **NB Model (wrong model):**

$$y_i \sim \text{NB}(\mu_i, \gamma),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^{5} a_j Z_{i,j};$$

Subsection 2

Z-residual Diagnostics

Posterior RSP vs. Posterior MSP Z-residual: Scatter Plot

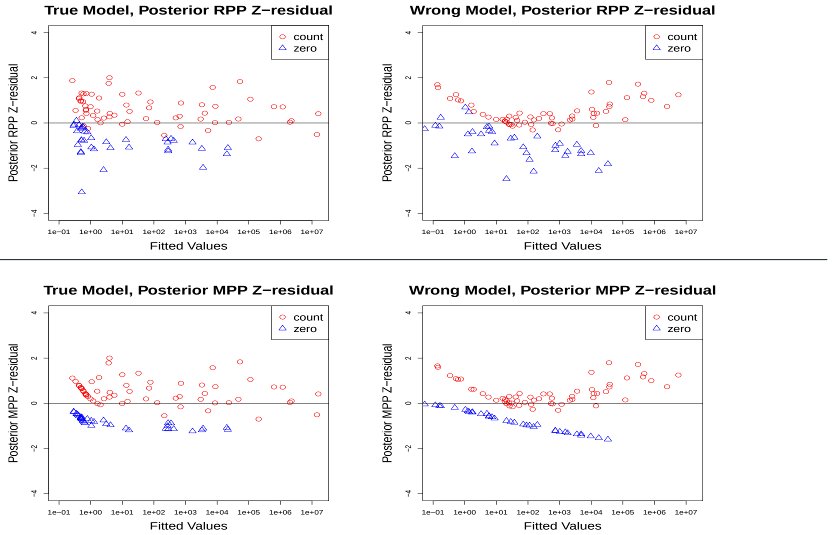


Figure 1: Posterior RSP Z-residuals and posterior MSP Z-residuals vs. covariate for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Posterior RSP vs. Posterior MSP Z-residual: Q-Q Plot

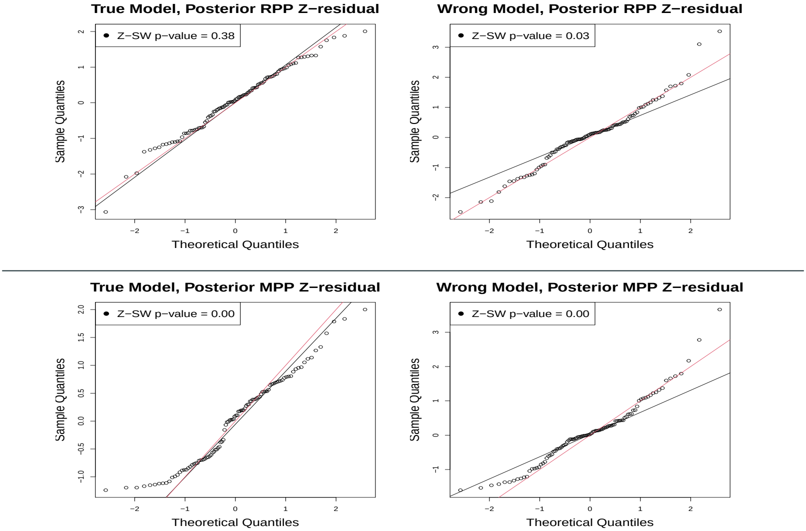


Figure 2: Q-Q plots for posterior RSP Z-residuals vs. posterior MSP Z-residuals for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Subsection 3

Replicated Studies for Estimating Type-I Error Rates and Powers

The Distributions of the RSP SW P-values

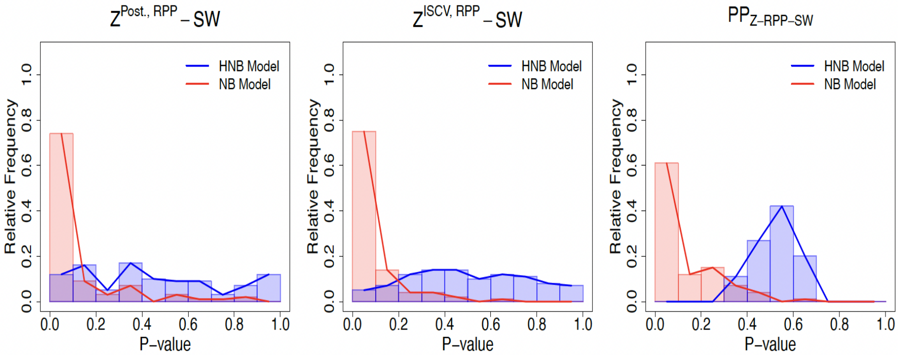


Figure 3: Histograms of P-values based on Shapiro-Wilk (SW) tests of posterior RSP Z-residuals vs. ISCV RSP Z-residuals vs. posterior predictive RSP Z-residuals when the dataset is simulated from HNB. The number of replications is 100. The zero proportion is $p = 30\%$.

The Distributions of the MSP SW P-values

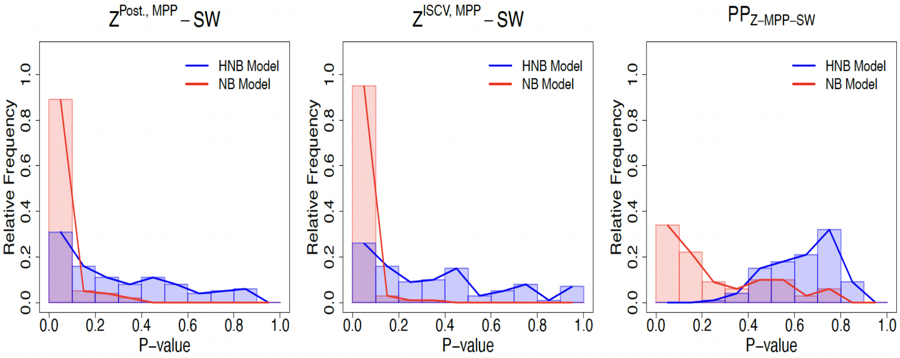


Figure 4: Histograms of P-values based on Shapiro-Wilk (SW) tests of posterior MSP Z-residuals vs. ISCV MSP Z-residuals vs. posterior predictive MSP Z-residuals when the dataset is simulated from HNB. The number of replications is 100. The zero proportion is $p = 30\%$.

The Distributions of the χ^2 P-values

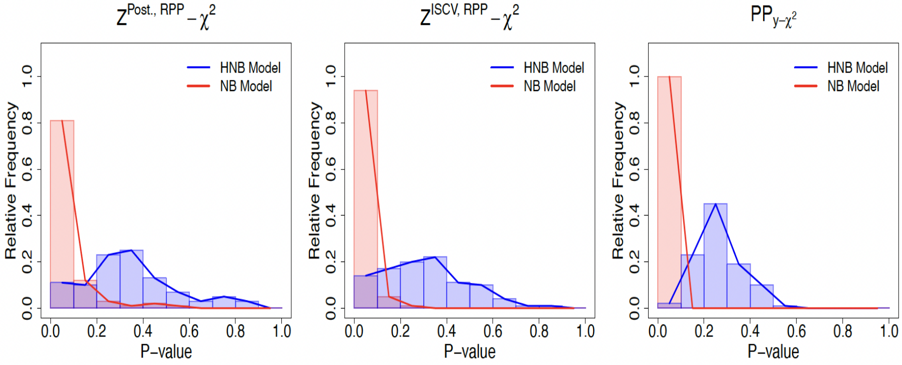


Figure 5: Histograms of P-values based on χ^2 tests of posterior RSP Z-residuals vs. ISCV RSP Z-residuals vs. posterior predictive χ^2 when the dataset is simulated from HNB. The number of replications is 100. The zero proportion is $p = 30\%$.

Model Rejection Rates

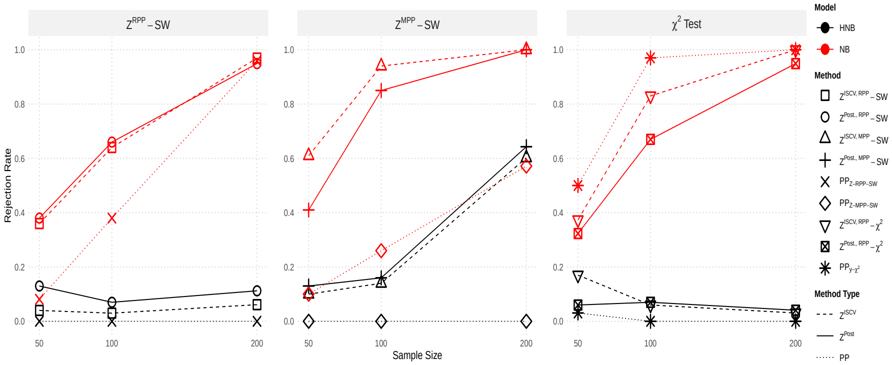


Figure 6: Model rejection rates based on Shapiro-Wilk (SW) tests of RSP vs. MSP Z-residuals when the dataset is simulated from HNB. A model is rejected when the test p-value is smaller than 5%. The number of replications is 100. The zero proportion is $p = 30\%$.

Section 6

Simulation Study II:
Detection of Wrong Covariate Functional Forms

Subsection 1

Data Generation and Model Fitting

Data Generation and Model Fitting

We add the **logarithmic covariate effect** to $X_{i,1}$ in the count model.

- Two covariates: $X_{i,1} \sim \text{Unif}(0, 15)$, $X_{i,2} \sim \text{Binom}(1, 0.5)$.
- Five noises: $Z_{i,j} \sim N(0, 1), j = 1, \ldots, 5$.
- Regression coefficients:
  $\alpha_0 = -1, \alpha_1 = 6, \alpha_2 = 1, \beta_0 = -1, \beta_1 = -1, \beta_2 = -5$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2}$.
- Zero model: $\text{logit}(\pi_i^o) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2}$.
- The zero indicator: $O_i \sim \text{Binom}(1, \pi_i^o)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
  where $C_i \sim \text{Truncated-NB}(\mu_i, \gamma, \pi), \gamma = 4$ is the dispersion parameter,
  and $\pi = \gamma/(\gamma + \mu_i)$ is the probability.
- The steps are repeated for all simulated datasets. We generate 600 response variables from the logarithmic HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$.

Data Generation and Model Fitting (cont'd)

- **Logarithmic HNB Model (true model):**

$$y_i \sim \mathrm{HNB}(\mu_i, \gamma, \pi_i^o),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2} + \sum_{j=1}^{5} a_j Z_{i,j},$$

$$\mathrm{logit}(\pi_i^o) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^{5} b_j Z_{i,j};$$

- **Linear HNB Model (wrong model):**

$$y_i \sim \mathrm{HNB}(\mu_i, \gamma, \pi_i^o),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_{i,1} + \alpha_2 X_{i,2} + \sum_{j=1}^{5} a_j Z_{i,j},$$

$$\mathrm{logit}(\pi_i^o) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^{5} b_j Z_{i,j}.$$

Subsection 2

Z-residual Diagnostics

Posterior RSP Z-residual: Scatter Plot

Figure 7: Posterior RSP Z-residuals vs. covariate $X_{i,1}$ for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100$, $p = 30\%$).

Posterior RSP Z-residual: Q-Q Plot

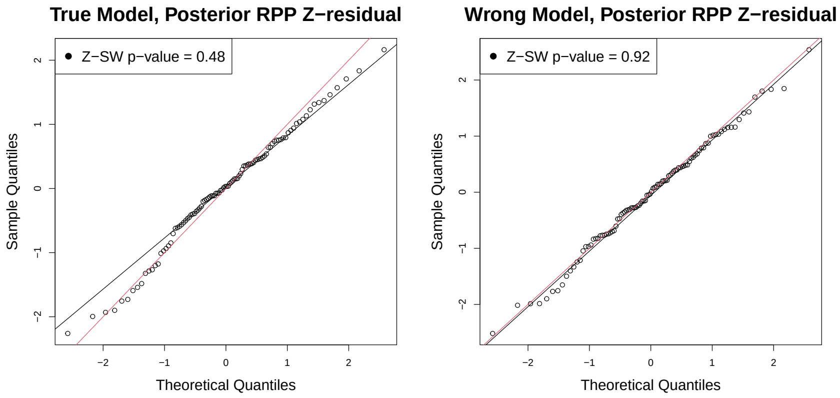
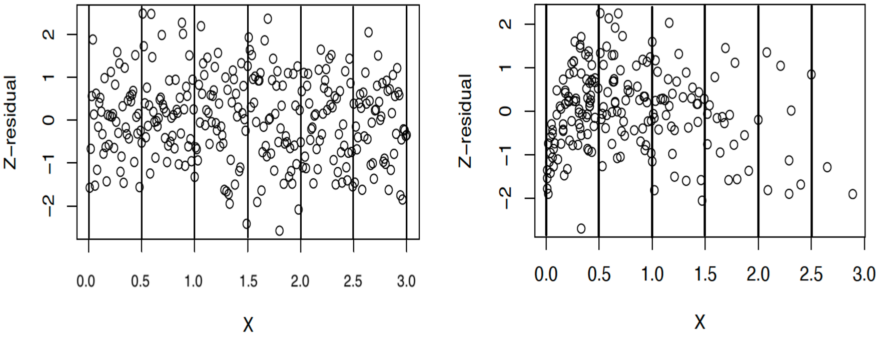


Figure 8: Q-Q plots for posterior RSP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100$, $p = 30\%$).

ANOVA with Partitioned Z-residuals for Capturing Mean Trend

Figure 9: An illustrative plot showing how to construct the non-homogeneity test with Z-residuals: dividing Z-residuals by a covariate or linear predictor (LP) with equally-spaced interval, then testing the equality of the means of grouped residuals. We apply the F-test in ANOVA to test the equality of the means of grouped Z-residuals.



Box Plot for ANOVA Tests

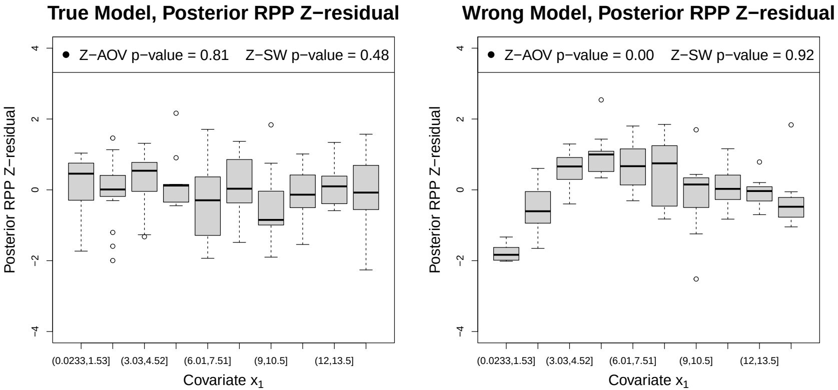


Figure 10: Box plots of posterior RSP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$). The p-values for the SW test and ANOVA test are shown on the top.

Subsection 3

Replicated Studies for Estimating Type-I Error Rates and Powers

What We Are Comparing

- SW vs. ANOVA Tests Applied to RSP Z-residuals
- Z-residuals vs. Posterior Predictive Checks

The Distributions of the RSP SW P-values

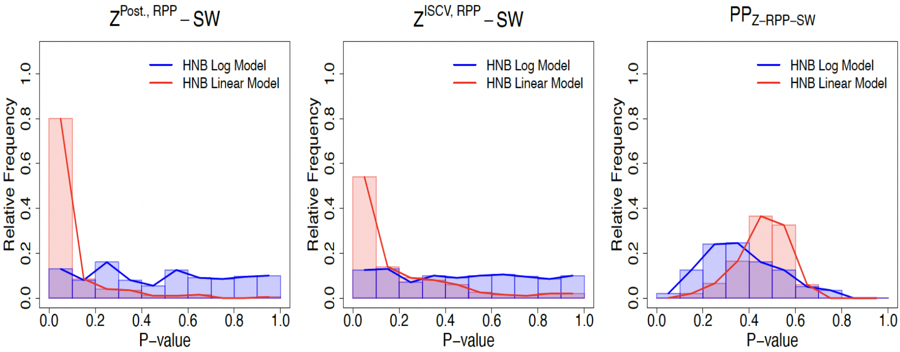


Figure 11: Histograms of P-values based on Shapiro-Wilk (SW) tests of posterior RSP Z-residuals vs. ISCV RSP Z-residuals vs. posterior predictive RSP Z-residuals when the dataset is simulated from HNB. The number of replications is 200. The zero proportion is $p = 30\%$.

The Distributions of the RSP AOV P-values

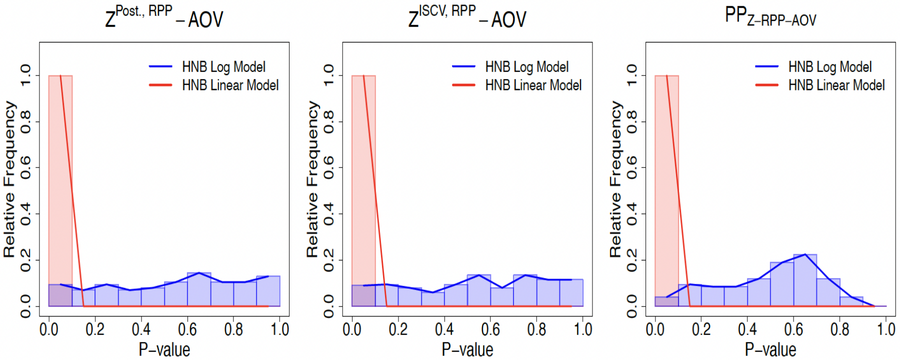


Figure 12: Histograms of P-values based on ANOVA tests of posterior RSP Z-residuals vs. ISCV RSP Z-residuals vs. posterior predictive RSP Z-residuals when the dataset is simulated from HNB. The number of replications is 200. The zero proportion is $p = 30\%$.

The Distributions of the χ^2 P-values

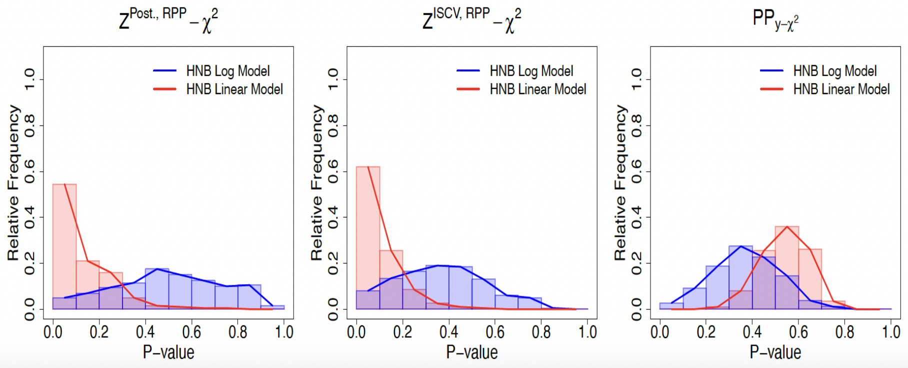


Figure 13: Histograms of P-values based on χ^2 tests of posterior RSP Z-residuals vs. ISCV RSP Z-residuals vs. posterior predictive χ^2 when the dataset is simulated from HNB. The number of replications is 100. The zero proportion is $p = 30\%$.

Model Rejection Rates

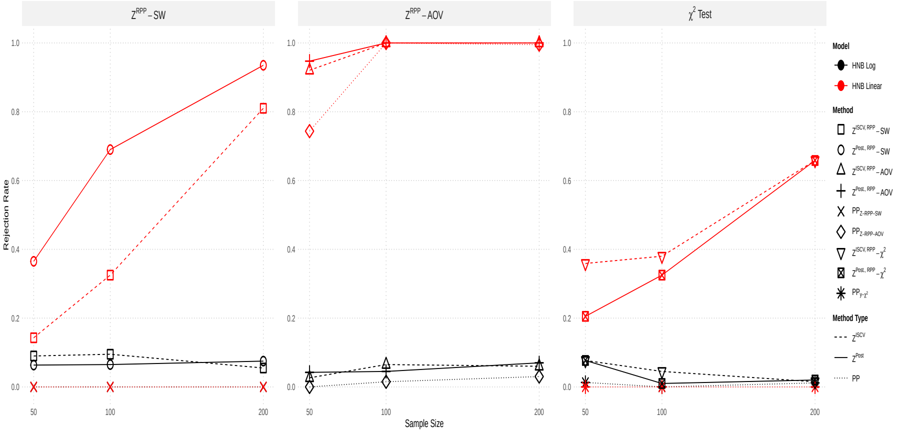


Figure 14: Model rejection rates of RSP Z-residual-based SW (Z-SW) Tests versus RSP Z-residual-based ANOVA (Z-ANOVA) Tests when the dataset is simulated from logarithmic HNB. A model is rejected when the test p-value is smaller than 5%. The number of replications is 600. The zero proportion is $p = 30\%$.

Section 7

Real Data Application

Data Descriptions

- The data was collected from the National Medical Expenditure Survey (NMES), conducted during 1987 and 1988.
- The dataset analyzed in this study focuses on 4,406 elderly individuals in the United States regarding their healthcare demands.
- The response variable under study is the number of emergency department (ED) visits.
- There are 18 covariates, including demographic factors, socioeconomic variables, and health metrics.

Data Analysis

- We fitted hurdle negative binomial (HNB), negative binomial (NB), hurdle Poisson (HP), and Poisson regression models.
- From the previous work, models are reduced to following covariates:
    - Number of chronic conditions (*numchron*)
    - Self-perceived health (*health*, excellent vs. poor; average vs. poor)
    - Limited activities of daily living (*adldiff*, yes vs. no)
    - Number of years of education (*school*)
- The black race (*black*) was significantly associated with increased ED use in the Poisson and HP models but not in the HNB and NB models.

Data Analysis (cont'd)

- **HNB Model:**

$$
\begin{aligned}
y_i &\sim \mathrm{HNB}(\mu_i, \gamma, \pi_i^o), \\
\log(\mu_i) &= \alpha_0 + \alpha_1 * \mathit{numchron}_i + \alpha_2 * \mathit{health}_i \\
&\quad + \alpha_3 * \mathit{adldiff}_i + \alpha_4 * \mathit{school}_i, \\
\mathrm{logit}(\pi_i^o) &= \beta_0 + \beta_1 * \mathit{numchron}_i + \beta_2 * \mathit{health}_i \\
&\quad + \beta_3 * \mathit{adldiff}_i + \beta_4 * \mathit{school}_i;
\end{aligned}
$$

- **NB Model:**

$$
\begin{aligned}
y_i &\sim \mathrm{NB}(\mu_i, \gamma), \\
\log(\mu_i) &= \alpha_0 + \alpha_1 * \mathit{numchron}_i + \alpha_2 * \mathit{health}_i \\
&\quad + \alpha_3 * \mathit{adldiff}_i + \alpha_4 * \mathit{school}_i;
\end{aligned}
$$

Data Analysis (cont'd)

- **HP Model:**

$$
\begin{aligned}
y_i &\sim \mathrm{HP}(\mu_i, \pi_i^o), \\
\log(\mu_i) &= \alpha_0 + \alpha_1 * \mathit{numchron}_i + \alpha_2 * \mathit{health}_i + \alpha_3 * \mathit{adldiff}_i \\
&\quad + \alpha_4 * \mathit{school}_i + \alpha_5 * \mathit{black}_i, \\
\mathrm{logit}(\pi_i^o) &= \beta_0 + \beta_1 * \mathit{numchron}_i + \beta_2 * \mathit{health}_i + \beta_3 * \mathit{adldiff}_i \\
&\quad + \beta_4 * \mathit{school}_i + \beta_5 * \mathit{black}_i;
\end{aligned}
$$

- **Poisson Model:**

$$
\begin{aligned}
y_i &\sim \mathrm{Poisson}(\mu_i), \\
\log(\mu_i) &= \alpha_0 + \alpha_1 * \mathit{numchron}_i + \alpha_2 * \mathit{health}_i + \alpha_3 * \mathit{adldiff}_i \\
&\quad + \alpha_4 * \mathit{school}_i + \alpha_5 * \mathit{black}_i.
\end{aligned}
$$

Posterior RSP Z-residuals: Q-Q Plots

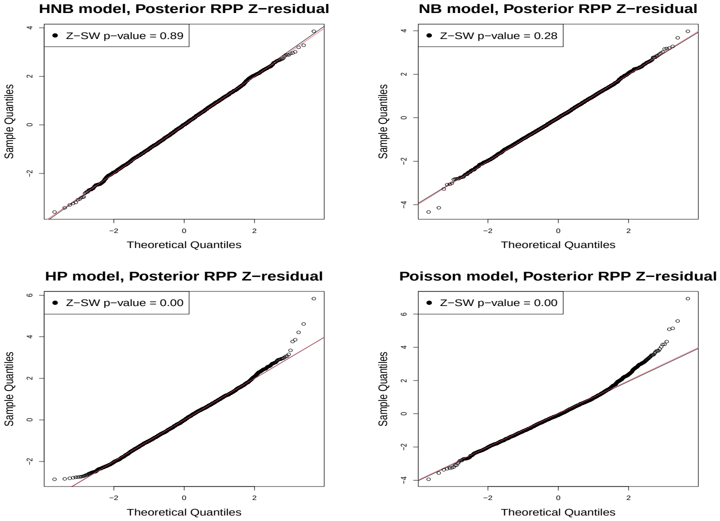
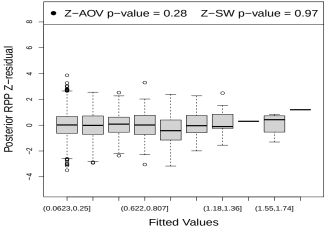
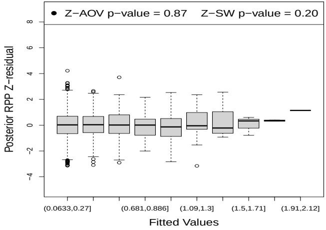


Figure 15: Q-Q plots for Posterior RSP Z-residuals for HNB, NB, HP, and Poisson models.

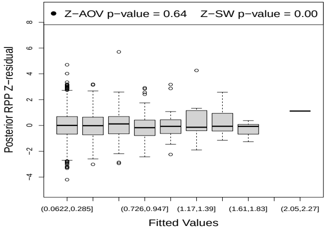
Posterior RSP Z-residuals: Box Plots



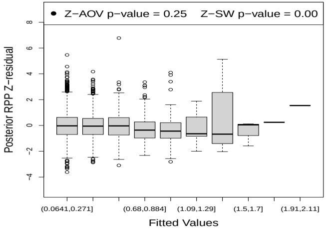
(a) HNB



(b) NB



(c) HP



(d) Poisson

Section 8

Conclusions, Future Work, and References

Conclusions and Future Work

- Z-residuals are well-calibrated, powerful, and informative for diagnosing Bayesian models.
- The Z-residual diagnostic methods can be significantly more powerful than posterior predictive p-values.
- The randomization in the RSP is necessary to produce well-calibrated diagnostic p-values based on Z-residuals.
- The ISCV Z-residuals may or may not be better than the posterior Z-residuals.
- Much more research work is needed to further extend and enhance the application of Z-residuals for other misfits, eg.,
  (1)detection of over-fitting, (2) implementation errors, (3) models with multiple outcome variables, (4) Spatial-temporal dependency, and many more

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